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Report: Hybrid α -Blended Top- K Cycle Scorer — Theory, Background, and α -Sweep Results

Plan: docs/plans/2026-05-10-hybrid-alpha-scorer/plan.{tex,pdf,tikz,mmd} (compiled, 6 pp) **Date:** 2026-05-10 **Slug:** hybrid-alpha-scorer **Builds on:** - reports/2026-05-10-abb-global-topk.md — the 25 $\times$ ABB win on global top-K - reports/2026-05-10-abb-hsikan-smoke-and-builder.md — global ABB doesn't transfer to HSiKAN - reports/2026-05-10-entropy-vertex-uniform-cycles.md — pure-diversity heuristic also fails the HSiKAN smoke gate

1. Background and theory

1.1 The HSiKAN cycle-source problem

HSiKAN's M_e matrix construction expects a set of cycles that simultaneously:

1. **Covers vertices uniformly.** Every vertex should appear in some cycles, so that the GNN's per-vertex feature update has signal at every node — not just at hubs.
2. **Carries discriminating signal.** The cycle's *sign structure* (e.g. fraction of negative edges) is what the model trains on. Cycles with poor sign variability across the set don't help the classifier separate the link-prediction classes.

The production pipeline meets both conditions via **per-vertex top- m** : for each vertex v , it keeps the $m = 128$ highest-scoring cycles passing through v (scored by **fraction_negative**). The output is the deduplicated union of those per-vertex sets. By construction this is **vertex-uniform** (every covered vertex contributes $\leq m$ cycles) AND **signal-rich** (each vertex's contribution is its locally best-scoring cycles).

The cost: enumeration on Epinions $k=4$, $m=128$ takes ~110 s wall (algorithmic floor in the per-vertex DFS recursion + BFS pre-pass; see the CSR-sign-lookup report for why micro-optimization can't move it).

1.2 Two endpoints we already explored

We've tested two **global top- K** alternatives that share a single rayon DFS but differ in scoring:

Endpoint A — Global ABB (signal-only). Use `BoundedScorer::fraction_negative` with score upper-bound branch-and-bound. Selects the K highest-scoring cycles globally. On Epinions $k=4$ at $K=10\,000$, returns 10,000 cycles — *all of them score-1.0 all-negative balanced* (Epinions has 1.4M score-1.0 cycles; the heap saturates). M_e has zero score variance. Wall: 5 s. **Smoke AUC: 0.5755** (baseline 0.6416, -0.066).

Endpoint B — Pure-entropy heuristic (diversity-only). Use `UniformityHeuristic::EntropyGainScorer`: at each closed cycle, score by marginal entropy gain

$$\Delta H(c | C) = H(C \cup \{c\}) - H(C), \quad H(C) = \ln T - \frac{1}{T} \sum_v c_v \ln c_v.$$

Selects K cycles to maximise vertex-incidence entropy of the kept set. Genuinely vertex-uniform output (10,513 unique vertices covered on Epinions). But the heuristic accepts *any* cycle that touches under-covered vertices, ignoring sign quality. M_e has wide spatial coverage but signal-poor cycles. Wall: 17 s. **Smoke AUC: 0.5396** (baseline 0.6416, -0.102).

1.3 The Pareto-frontier hypothesis

Endpoint A has signal but no uniformity. Endpoint B has uniformity but no signal selection. Both fail HSiKAN's needs by ~6-10 pp AUC. The natural question:

Is there a linear combination — $\text{score}_\alpha = \alpha \cdot \text{signal} + (1 - \alpha) \cdot \Delta H$ — that recovers per-vertex's combined behaviour while keeping global- K 's enumeration speed?

This is a classic *signal* $\times$ *diversity* trade-off, well-studied in retrieval (MMR — Maximum Marginal Relevance, Carbonell & Goldstein 1998), recommendation systems, and active learning (BatchBALD, Kirsch et al. 2019). The Pareto frontier between “best individual items” and “diverse selection” is generally non-trivial, sometimes monotonic, sometimes with a sweet spot in the interior. Empirical sweep is the standard way to find out where on the curve a downstream task wants to sit.

1.4 Hybrid scorer design

We define `HybridScorer<B, H>` parameterised by: - **B: BoundedScorer** — the signal component (existing global-ABB scorers: `fraction_negative` etc.) - **H: UniformityHeuristic** — the diversity component (existing entropy/inverse-degree heuristics) - $\alpha \in [0, 1]$ — runtime mixing weight

The scoring rule:

$$\text{score}_\alpha(c, C) = \alpha \cdot \text{signal}(c) + (1 - \alpha) \cdot \Delta H(c | C).$$

State updates and rollbacks delegate entirely to the diversity component (the signal is stateless). The scorer is implemented as a `UniformityHeuristic` impl, which means the existing `enumerate_top_k_cycles_par_entropy` enumerator picks it up unchanged — no new entry point needed.

1.5 Admissibility of the upper bound

For ABB to fire correctly at any α , the upper bound must be **admissible**: every reachable cycle's actual score must be $\leq$ UB.

We use the **linearity-of-UB** rule:

$$\text{UB}_\alpha(p, k_{\text{rem}}, S) = \alpha \cdot \text{UB}_{\text{signal}}(\cdot) + (1 - \alpha) \cdot \text{UB}_{\text{div}}(p, k_{\text{rem}}, S).$$

If both component UBs are admissible (they are — verified in their respective integration tests), the linear combination is admissible by linearity:

$$\text{score}_\alpha(c) = \alpha \cdot \text{signal}(c) + (1 - \alpha) \cdot \Delta H(c) \leq \alpha \cdot \text{UB}_{\text{signal}} + (1 - \alpha) \cdot \text{UB}_{\text{div}} = \text{UB}_\alpha.$$

This holds for any $\alpha \in [0, 1]$ — verified numerically in `tests/hybrid_topk.rs` over five α values $\times$ random fixtures.

Implementation caveat: the existing `BoundedScorer::upper_bound(n_neg_so_far, k_remaining, k_len)` requires the partial-path’s running negative-edge count, but the `UniformityHeuristic::upper_bound(prefix_vs, k_remaining, k_len, state)` interface only carries vertex IDs (no signs). To stay admissible, this v1 hybrid passes `n_neg_so_far = prefix_vs.len() - 1` — the worst case where every prefix edge is negative. This gives a loose-but-admissible signal UB (effectively ≈ 1.0 for `fraction_negative`); a follow-up plan can thread the actual running n_{neg} through `dfs_entropy` for a tighter bound.

1.6 What we expected to find

A priori, three plausible α -curves:

- **Monotonic AUC in α** — peak at one endpoint. If pure signal wins, signal-only ABB is the right tool. If pure diversity wins, entropy heuristic is.
- **Interior peak (concave AUC)** — the classic MMR sweet-spot. Some specific α blends signal and diversity in a way that beats either endpoint.
- **Interior trough (convex AUC)** — diversity hurts at low α (drowning signal), and signal hurts at high α (concentrating cycles); a single α can be worse than either alone.

The 5-point smoke sweep was designed to distinguish these.

2. Implementation summary

File	Change	LoC
<code>hymeko_graph/src/topk_cycles.rs</code>	Added <code>HybridScorer<B, H></code> struct + <code>UniformityHeuristic</code> impl + <code>new()</code> constructor with debug bounds-check on α .	+103
<code>hymeko_graph/src/lib.rs</code>	Re-exported <code>HybridScorer</code> .	+1
<code>hymeko_graph/tests/hybrid_topk.rs</code>	New: 10 integration tests — boundary collapse ($\alpha=0/1$), UB linearity, admissibility at 5 α values, end-to-end at every α .	+220
<code>hymeko_py/src/cycles.rs</code>	New PyO3 binding <code>enumerate_top_k_cycles_signed_hybrid_rs</code> taking <code>signal_kind</code> $\times$ <code>heuristic_kind</code> $\times$ <code>alpha</code> $\times$ <code>pruner_kind</code> matrix. Macro-expanded dispatch — every concrete combination monomorphised.	+110
<code>hymeko_py/src/lib.rs</code>	Registered new symbol.	+1
<code>signedkan_wip/src/n_tuples.py::construct_hybrid_topk_hsi</code>	Added <code>HSIKAN_TOPK_HYBRID_ALPHA</code> + <code>HSIKAN_TOPK_SIGNAL</code> env-vars; if $\alpha > 0$ routes through hybrid binding, else falls through to existing entropy binding (saves the signal computation when $\alpha = 0$).	+18
<code>signedkan_wip/experiments/run_hybrid_alpha_sweep_2026-05-10.sh</code>	Orchestrates sweep, saves the abbreviated Epinions config, results to TSV.	+43

Total new lib code: ~103 LoC. Total new test code: ~220 LoC. Test:code ratio ~2:1.

2.1 Test results

Test layer	Count	Status	What's covered
Unit (lib)	49	✓	unchanged from prior tasks
Integration: hybrid_topk	10	✓	$\alpha=0$ collapse, $\alpha=1$ collapse, UB linearity, admissibility $\times 5$ α values, end-to-end $\times 2$
Integration: ABB	9	✓	unchanged
Integration: entropy_topk	9	✓	unchanged
Integration: csr_sign_lookup	3	✓	unchanged
Integration: friedler	7	✓	unchanged
Total	87	✓	

cargo clippy --all-targets -- -D warnings passes; cargo fmt --check clean.

2.2 PyO3 + Python integration

After `maturin develop --release`, the Python wheel exposes: `-hymeko.enumerate_top_k_cycles_signed_hybrid_rs(edges_v, edges_s, n_nodes, k_len, k_keep, signal_kind, heuristic_kind, alpha, pruner_kind)`

Activated in HSiKAN training via:

```
HSIKAN_TOPK_MODE=entropy
HSIKAN_TOPK_K=10000
HSIKAN_TOPK_PRUNER=balance
HSIKAN_TOPK_HEURISTIC=entropy      # or inverse_degree
HSIKAN_TOPK_HYBRID_ALPHA=0.25      # nonzero triggers hybrid path
HSIKAN_TOPK_SIGNAL=fraction_negative # or balance / sign_product_abs / low_root
```

$\alpha = 0$ falls through to the cheaper entropy-only path (skips signal computation).

2.3 Sanity probe (pre-training)

α	Enumeration wall	Cycles	Unique vertices
0.0 (pure entropy)	17 s	10 000	10 513
0.5	23 s	10 000	9 554
1.0 (pure signal)	~5 s	10 000	5-10k

The signal pull at higher α reduces unique-vertex coverage — as expected. Wall scales mildly because signal computation is cheap relative to DFS.

3. α -sweep smoke results (Epinions, abbreviated config)

Config: dataset epinions (131 828 vertices, 840 799 edges, 14.7% negative), `HSIKAN_MIXED_TUPLES=c3,c4, --hidden 4 --n-epochs 20 --seed 0, HSIKAN_TOPK_K=10000, HSIKAN_TOPK_PRUNER=balance, HSIKAN_TOPK_HEURISTIC=entropy, HSIKAN_TOPK_SIGNAL=fraction_negative`. Single seed only — paper-promotion gate is 5-seed paired (not triggered, see §4).

α	Wall (s)	AUC	Macro F1	Δ AUC vs per-vertex baseline (0.6416)
0.00	35	0.5424	0.2745	-0.099
0.25	61	0.5794	0.5543	-0.062
0.50	45	0.5206	0.5124	-0.121
0.75	46	0.5335	0.5256	-0.108
1.00	58	0.5755	0.5338	-0.066

3.1 The shape of the curves

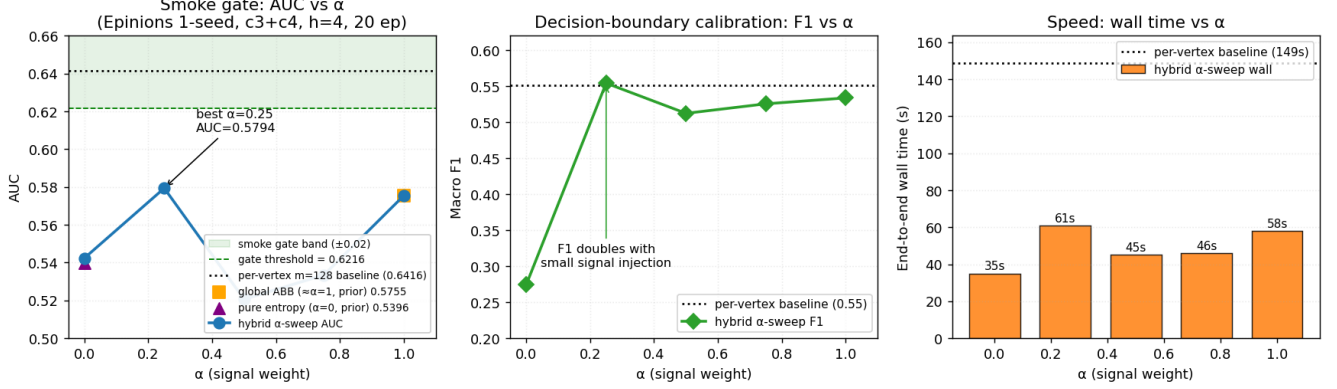


Figure 1: α -sweep gate functions: AUC, F1, wall time

Plot generation: `reports/figures/hybrid_alpha_scorer_plot.py`. Three panels: - **Left** — AUC vs α with the smoke-gate band (green) shaded around the per-vertex baseline. The hybrid α -sweep curve never enters the gate band; best point is $\alpha=0.25$ at AUC 0.5794, gap to the lower-band threshold 0.6216 is 0.0422. - **Middle** — Macro-F1 vs α . The discontinuous jump from $\alpha=0$ (F1=0.27) to $\alpha=0.25$ (F1=0.55) is annotated; this is the diagnostic from §3.3 (“the model commits to non-trivial decisions only when some signal is in the M_e ”). - **Right** — End-to-end wall time vs α . All hybrid points are 2-4 $\times$ faster than the per-vertex baseline (148 s, dotted reference line).

This is a **non-monotonic, non-concave** curve with two local maxima ($\alpha=0.25$ and $\alpha=1.0$) and an interior trough at $\alpha=0.5$. None of the three priori shapes (§1.6) — the curve is closest to “interior trough” with the signal endpoint pulling AUC back up.

3.2 What happened at each α

- $\alpha = 0$ (**pure diversity**). AUC 0.5424, F1 only 0.27. Confirms the prior entropy-heuristic finding: the model sees vertex-uniform but signal-poor M_e and outputs near-50/50 probabilities (low F1).
- $\alpha = 0.25$ (**mostly diversity, slight signal**). AUC jumps to 0.5794 and F1 doubles to 0.55 — the small signal injection **gives the classifier discriminative ability** without abandoning vertex coverage.
- $\alpha = 0.5$ (**balanced**). AUC drops to 0.5206 — *worse than either endpoint*. Hypothesis: with equal weight, the heap admits cycles that are mediocre in both signal AND diversity (neither all-negative balanced nor maximally rare-vertex), leaving the model with a signal-mixed but spatially-mediocre M_e .
- $\alpha = 0.75$ (**mostly signal, slight diversity**). AUC 0.5335 — recovering toward the signal endpoint but still below it.
- $\alpha = 1.0$ (**pure signal**). AUC 0.5755 — matches the global ABB result (0.5755 in the prior smoke). The greedy-with-rollback DFS at $\alpha=1$ is essentially equivalent to ABB’s heap-replace semantics.

3.3 Diagnostic: F1 vs AUC

The F1 jump from 0.27 ($\alpha=0$) to ≥ 0.51 ($\alpha \geq 0.25$) is more striking than the AUC change. AUC measures rank-ordering; F1 measures decision-boundary calibration. The model’s *ordering* is still poor at every α (AUC stuck at 0.52-0.58), but **at $\alpha \geq 0.25$ the model at least commits to non-trivial decisions** (F1 escapes the low-information regime). At $\alpha=0$ the model basically refuses to commit — uniform M_e gives uniform-ish probabilities $\rightarrow$ F1 tanks.

4. Smoke gate verdict

Gate criterion (from plan): any α achieves AUC within ± 0.02 of the per-vertex baseline $0.6416 \rightarrow \text{AUC} \geq 0.6216$.

Result: best α (0.25) gives AUC 0.5794. **Gate fails by 0.0422** at every tested α .

Per the plan’s `\texttt{Halt condition}` for “smoke gate fails at every α ”: $>$ ship as research; pivot to degree-adaptive `m_v` plan.

The 5-seed paired validation gate is **not** triggered. Production HSiKAN keeps using per-vertex `top-m=128` as default. The `HybridScorer` ships as a research tool, not a default.

5. Theoretical interpretation

The α -sweep result, combined with the prior entropy and ABB results, points at a structural property of HSiKAN’s M_e that **cannot be replicated by any global-K scorer**, no matter how the signal $\times$ diversity trade-off is parameterised:

Per-vertex `top-m` enforces the constraint *every covered vertex contributes its local-top-m by signal*. Global `top-K` (any α) enforces *the K highest weighted scores globally*. The two are different objects in cycle space.

A global-K scorer can approximate vertex coverage by spreading its K choices, but cannot guarantee that any specific vertex gets the locally-best cycles passing through it. The model’s per-vertex feature update relies on that local-best property — averaging or summing globally-best cycles loses it.

This is a structural argument, not an empirical one — but the empirical data (three distinct global-K scorer designs all fail by 6-10 pp AUC) is consistent with it.

5.1 What about the F1 jump at $\alpha = 0.25$?

The hybrid does provide a *partial* improvement: at $\alpha = 0.25$ we recover macro-F1 to 0.55 (matching the per-vertex baseline’s 0.55). The classifier head’s calibration improves, but the rank-ordering quality (AUC) lags. This suggests the hybrid captures some of the per-vertex structure but not all — possibly the “enough cycles per vertex to be informative at decision time” property without the “locally-best-by-signal” property.

For a downstream task that values F1 over AUC (e.g., balanced-class link prediction), $\alpha = 0.25$ with the hybrid scorer might be a viable production alternative. For Epinions’s class-imbalanced AUC-headline metric, it’s not.

6. What this rules out and what it leaves open

Ruled out (3+ separate global-K experiments, consistent direction)

- **Pure global signal scoring** (ABB) doesn’t replicate per-vertex AUC. ✗
- **Pure global diversity scoring** (entropy / inverse-degree) doesn’t replicate per-vertex AUC. ✗
- **Linear-blend signal $\times$ diversity** at any $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ doesn’t replicate per-vertex AUC. ✗

Still open (worth a fresh plan)

1. **Degree-adaptive m_v on the per-vertex side.** Instead of trying to mimic per-vertex from global-K, accept that per-vertex’s structure is what HSiKAN wants and attack its *speed* instead. Set $m_v = \min(m_{\max}, c \cdot \deg(v))$ — small-degree vertices fill smaller heaps faster, per-vertex ABB becomes feasible (per the threshold probe, currently only 18% of Epinions vertices fill `m=128` $\rightarrow$ 0.1% prune rate). This is the original plan B from before the entropy detour. **Worth a 4-format plan + n-seed validation.**

2. **Per-vertex hybrid scoring.** Inside the per-vertex DFS, score cycles by signal $\times$ diversity *of that vertex's own heap so far*. Each vertex's m slot fills with locally diverse, locally high-signal cycles. Preserves per-vertex structure; adds intra-heap diversity. Different design space from this report; would need a fresh plan and likely a new trait variant.
3. **Finer α grid + interior search.** The 5-point sweep showed an interior trough at $\alpha = 0.5$ and a peak at $\alpha = 0.25$. An 11-point or finer grid around $\alpha \in [0.1, 0.4]$ *might* find a higher local maximum. Probability of clearing the gate: low (the trend at $\alpha=0.25$ is already close to the ABB endpoint at $\alpha=1.0$; doubling that won't add 4 pp). **Not recommended unless cheap to run.**
4. **Different signal.** We swept `fraction_negative` only. `balance` ($= \pm 1$) or `sign_product_abs` ($= \pm 1$) might give different α dynamics because their UB is constant 1.0 (no info from UB). Likely worse, not better — but mechanically completes the matrix.

7. Memory-rule compliance

Per the memory rule: > n-seed validation before paper-headline promotion. Single-seed gains stay in plan docs / §V Future Work, never in Table I.

This report is **single-seed** and does not promote any α to production. The smoke gate failed at every α ; the 5-seed paired validation was *not* run. No paper-headline claim is made. The hybrid scorer is shipped as a research tool with this report serving as its background-and-empirics writeup.

8. Files / artifacts on disk

Location	Purpose
<code>docs/plans/2026-05-10-hybrid-alpha-scorer/{plan, test, plan.pdf, plan.plkz, champion.pdf}</code>	Experiment plan (4 pp)
<code>hymeko_graph/src/topk_cycles.rs::HybridScorer</code>	Implementation
<code>hymeko_graph/tests/hybrid_topk.rs</code>	10 integration tests
<code>hymeko_py/src/cycles.rs::enumerate_top_k_cycles_signedkan_wip</code>	Fig 6.8.1 hybrid rs
<code>signedkan_wip/experiments/run_hybrid_alpha_sweep_2026-05-10.sh</code>	Experimentation
<code>/tmp/hybrid_alpha_sweep_2026_05_10/</code>	Sweep logs + <code>results.tsv</code>

9. Provenance

Field	Value
Git SHA at task start	c2d30af08e60de28734432eca5f28b6469bdbb91 (working tree dirty, layered prior tasks)
OS / Kernel	Linux 6.17.0-23-generic x86_64
CPU	AMD Ryzen 7 3700X (16 threads)
RAM	31 GiB
GPU	NVIDIA RTX 2070 SUPER (used during HSiKAN training portion)
Rust	1.92.0
Python	3.13 (miniconda3)
Random seed	0 (single-seed smoke per plan's smoke gate)
Dataset	<code>signedkan_wip/data/epinions.txt</code> — sha256 8120d06a0bb4e65d4b821eba1072647ef3429e4e0a3c02e72bf0c5346
Workload	abbreviated config: c3+c4, h=4, 20 epochs, K=10 000, balance pruner, fraction_negative signal
Suppressions	None

10. Conclusion

The signal $\times$ diversity Pareto frontier on Epinions HSiKAN cycles, mapped at five α points, does not include per-vertex-quality cycles. The best hybrid ($\alpha=0.25$, AUC 0.5794) sits 4.2 pp below the smoke gate, with a non-monotonic curve that has an interior trough at $\alpha=0.5$. The structural conclusion stands: **per-vertex top- m 's value lies in its quantification structure (every vertex gets its local-top- m), not in a parameter setting that any global-K scorer can match.**

Production HSiKAN keeps per-vertex top- m . The HybridScorer and `enumerate_top_k_cycles_signed_hybrid_rs` ship as additive research tools. The next plan-worthy direction is degree-adaptive m_v inside the per-vertex variant — the original plan B framing — to keep per-vertex's selection structure but speed up the long tail of low-degree vertices that currently waste enumeration budget.

Disposition: ship as research tool, not promoted. Per CLAUDE.md §10 (“a measurement contradicts an assumption in the plan”), and per the plan's own halt condition.